

Futureproof

Equity Preservation Mortgage®

Modelling v14d (Optimized)

Reinsurance Review

LMI & Portfolio Risk Reinsurance Program: Recommended Structure and Interaction May 2026

Based on the Equity Preservation Mortgage (EPM) v14d (Optimized) Model
GBM (with Stochastic Drift) + Mean Reversion Equity Model (Shevchenko 2026)
50,000-path Monte Carlo Simulation | May 2026

Purpose: This paper sets out the recommended structure for the Futureproof Lenders Mortgage Insurance (LMI) program and its interaction with the Portfolio Risk Reinsurance program. It provides reinsurers with a comprehensive framework for structuring tail risk coverage for EPM portfolios.

***Important note on dollar amounts:** All dollar amounts in this paper are per-mortgage values for a representative \$1,500,000 property at 80% LVR. These amounts will vary proportionally for different home values, LVR ratios, and annuity levels. They are not global constants.*

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1. Executive Summary

The Equity Preservation Mortgage® is a strongly positive-expectation product, and — importantly — the homeowner bears none of its investment risk. The homeowner's income is contractually guaranteed regardless of market conditions; they retain their home throughout; and their home wealth (both original home equity and all property appreciation during the loan term) is preserved in every outcome (the product is non-recourse to the borrower).

The index-linked reference asset portfolio (held in the borrower's mortgage offset account) sits behind the guarantee — funding loan costs and its mortgage surplus or shortfall. Any end-of-term mortgage surplus arising from index appreciation over 30 years, accrues to the stakeholders carrying cashflow risk - Futureproof, the lender and funder, not the homeowner.

Those economics are sound too: across 50,000 simulated 30-year paths, **roughly 92% of mortgages fully self-fund** — the mortgage offset account grows past the loan balance, with a typical (mean) total surplus near \$1.3M (shared between the stakeholders) — and the two-layer insurance covers the minority tail.

This paper concerns that tail; the risk analysis that follows characterizes it, not the expected case.

For a reinsurer this is an attractive risk to write: a book where the great majority of mortgages self-fund in full, where any claim can only crystallize once (at loan maturity), and where loss severity is modest. The two-layer architecture below protects lenders and wholesale funders against the residual tail.

This paper addresses two key questions:

- **Part A:** How should the individual-mortgage LMI program settings interact with the loan-book / portfolio risk reinsurance program settings?
- **Part B:** How should a reinsurer structure the Portfolio Risk Reinsurance Program?

Key Findings — Per-Layer (\$1.5M home at 80% LVR)

Metric	LMI Layer	Tail Risk Reinsurance	Combined
Insurance scope	Pays on every deficit path	Pays only on severe-deficit paths (subset of LMI)	100% coverage from \$0
Covers deficits	Up to \$234,495 per mortgage	Excess above \$234,495 per mortgage	All deficits from \$0
Probability of Claim (% of all mortgages) ¹	8.37%	1.67% (subset of LMI)	8.37% (any insurer pays)
Fair Premium (PV)	\$10,168 per mortgage	\$1,827 per mortgage	\$11,995 per mortgage
Loaded Premium (50%)	\$15,253 per mortgage	\$2,740 per mortgage	\$17,993 per mortgage
Premium timing	Upfront at origination	Upfront at origination	Upfront at origination

¹ Reinsurance claims are a strict subset of LMI claims — every reinsurance claim is also an LMI claim, because LMI pays first-loss from \$0. The 1.67% reinsurance frequency is the ~20% subset of severe-deficit paths within the 8.37% total deficit probability. "Combined" is the probability that any insurer pays anything — equal to the LMI frequency, not the sum.

Outcome Distribution at Maturity

Every mortgage falls into exactly one of the following buckets at year 30. **The great majority finish in surplus with no claim**; the two insurance layers split the deficit *dollars* only on the minority tail — and both layers can pay on the same mortgage.

Year-30 outcome	Frequency (% of all mortgages)	LMI pays	Reinsurer pays
In surplus (no claim)	91.63%	\$0	\$0l
Moderate deficit ($\leq \$234,495$)	6.70%	Full deficit	\$0
Severe deficit ($> \$234,495$)	1.67%	\$234,495 (capped)	Excess
Any insurance claim	8.37%	—	—

Critical distinction: The EPM is a mortgage-based loan. Insurance claims can only arise at mortgage expiry (end of the 30-year term), not at any intermediate point. The insurance structure provides 100% coverage of all deficit outcomes from dollar zero — there is no deductible or uninsured gap. The two layers divide responsibility by loss severity: the LMI insurer covers smaller deficits, and the reinsurer covers the excess on larger deficits.

Parameter Uncertainty and Three-Scenario Framework

The premium numbers above are produced under the model's base-case parameter set (MLE central estimates from historical data, 0.046% collar cost, base correlation). A companion paper — *Model Assumptions & Parameter Risk* — stress-tests each parameter against academic benchmarks (Shiller CAPE, Jordà-Schularick-Taylor, Black-Scholes collar pricing) and defines three scenarios that reinsurers should use side-by-side when structuring coverage:

Scenario	PoD (Yr 30)	Tail-layer claim rate	Use in reinsurance structuring
Base (model-as-written)	8.37%	1.67%	Quoted fair premium; pricing floor
Realistic-central	40%	— (not separately run)	Premium anchor; treaty rate-on-line
Adverse-plausible	69%	— (not separately run)	Capital sizing; attachment/limit design; PML

Scenario PoD figures are *xlsm-verified* (realistic-central and adverse-plausible parameter sets, defined in the companion **Model Assumptions & Parameter Risk** paper). The tail-layer (reinsurance) claim rate was not separately simulated for each scenario; it remains structurally bounded by the per-mortgage reinsurance PoC ceiling and rises with PoD. The three-scenario logic (pricing floor / anchor / capital sizing) is the recommended structuring framework.

Reinsurer guidance: Structural features of the EPM (run-off mechanism, Payments Waterfall, ~10–12% loss severity given a deficit*, 100% insurance from dollar zero, ~30-year float on upfront premium) are scenario-robust) are structural features of the product, not parameter-dependent. What moves across stress scenarios is the expected loss frequency. Reinsurance pricing should anchor to the realistic-central scenario (PoD 40%), and capital and limit sizing to the adverse-plausible scenario (PoD 69%) — both PoD figures are *xlsm-verified* and detailed in the companion *Parameter Risk* paper.

2. Stochastic Model Selection and Calibration

The EPM requires robust stochastic modelling of two key risk drivers over a 30-year horizon: S&P; index returns and interest rates. This section describes the model selection process, explains why each model was chosen, and details the parameter estimation methodology.

Interest Rate Model: Vasicek/Ornstein-Uhlenbeck Process

Cash rates are modelled using the Vasicek (Ornstein-Uhlenbeck) process, the standard model for mean-reverting interest rates. The exact discretisation (Shevchenko 2026, Section 1) is:

$$r(t+1) = r(t) * w + \theta * (1 - w) + v * \epsilon(t)$$

where $w = \exp(-\kappa * dt)$, $v = \sigma * \sqrt{(1 - w^2) / (2 * \kappa)}$, and $\epsilon \sim N(0,1)$.

This model captures the empirically observed tendency of interest rates to revert to a long-run mean, which is essential for realistic 30-year projections. A random walk model would produce unrealistic rate paths over such a long horizon.

Equity Model: GBM (with Stochastic Drift) + Mean Reversion

The modelling of equities return of a particular stock or basket of stocks, may not be the appropriate methodology to apply to the modeling of index returns. The performance of an index is structurally different from the performance of individual stocks.

The stochastic modelling process *fundamentally changes* when you move from a single stock to a broad index like the S&P 500, because the underlying economic drivers, statistical properties, and sources of uncertainty are structurally different. A single stock's return distribution is dominated by idiosyncratic risk, valuation reversion, and firm-specific events — none of which survive diversification. An index's return distribution is dominated by macro regimes, volatility clustering, and cross-sectional averaging — which makes its stochastic behavior smoother, more stable, and structurally different.

A. The statistical structure is different

Single stock = fat tails + jumps + valuation sensitivity

A single stock exhibits:

- Large idiosyncratic jumps (earnings surprises, scandals, M&A, fraud, product failures)
- Mean reversion to valuation anchors (P/E, P/B, ROE, growth expectations)
- High volatility of volatility
- Non-stationary drift (firm lifecycle: growth → maturity → decline)
- Survivorship bias risk (delisting, bankruptcy)

These features produce:

- Heavy-tailed return distributions
- Frequent discontinuities
- Drift that is *not* stable over long horizons
- Volatility that is *not* well-behaved

Modelling implication: You need jump-diffusion, stochastic volatility, and valuation-linked drift (e.g., P/E reversion models).

B. Index = diversification smooths away idiosyncratic noise

The S&P 500 is:

- A market-cap weighted portfolio of 500 stocks
- Continuously rebalanced
- Continuously refreshed (dead firms removed, winners added)
- Dominated by macro factors, not firm-specific noise

This produces:

- Much lower jump intensity
- Much more stable drift
- Volatility clustering driven by macro regimes, not firm events
- Stationary long-run behaviour (ergodic properties emerge)
- Near-lognormal behaviour at short horizons, but fat-tailed at long horizons due to regime shifts

Modelling implication: You model the index with:

- Stochastic volatility (Heston, GARCH, HAR)
- Regime switching (expansion, slowdown, recession, recovery)
- Occasional macro jumps (crises), not firm-level jumps

C. Valuation dynamics behave differently

Single stock:

P/E matters directly- A stock's expected return is tightly linked to:

- Reversion of P/E to sector norms
- Changes in expected earnings growth
- Firm-specific discount rate changes

If P/E is high, expected returns fall; if P/E is low, expected returns rise. This is a **first-order driver** of drift.

Index:

P/E matters, but only at the aggregate level - Index valuation:

- Moves slowly
- Mean reverts over multi-year horizons
- Is influenced by macro rates, inflation, and risk premia
- Is *not* driven by any single firm's valuation

The index's drift is therefore:

- More stable
- Less sensitive to micro valuation
- More sensitive to macroeconomic regimes

D. Drift is fundamentally different

Single stock drift = firm lifecycle

A firm's drift changes as it:

- Grows
- Matures
- Declines
- Faces competition
- Innovates or fails

This makes drift **non-stationary**.

Index drift = macroeconomic growth + risk premium

The S&P 500's drift is driven by:

- Productivity growth
- Population growth
- Inflation
- Corporate margins
- Equity risk premium

This makes drift **much more stable** over decades.

E. Volatility behaves differently

Single stock volatility:

- High
- Unstable
- Firm-event driven
- Often mean-reverting but with large spikes

Index volatility:

- Lower
- Highly persistent
- Driven by macro regimes
- Exhibits long memory (HAR, GARCH-type behaviour)

This is why **Heston or regime-switching models** fit index behaviour far better than GBM.

F. Tail risk is different

Single stock:

- Bankruptcy risk
- Delisting
- Catastrophic firm-specific events
- Extreme negative skew

Index:

- No bankruptcy risk
- Tail events are macro (crises)
- Less skewed
- More symmetric distribution

This is why **jump intensity is much lower** for indices.

G. Rebalancing and survivorship bias change the maths

The S&P 500:

- Removes losers
- Adds winners
- Reweights toward outperformers

This creates:

- **Implicit momentum**
- **Reduced left-tail risk**
- **Higher long-run drift** than a static basket
- **Different stochastic properties** than any single constituent

A single stock has none of these protections.

I. Practical modelling consequences

Single stock modelling requires:

- Jump-diffusion
- Stochastic volatility
- Valuation-linked drift
- Possibly structural models (Merton credit-equity link)
- Non-stationary parameters

Index modelling requires:

- Stochastic volatility (Heston, GARCH, HAR)
- Regime switching
- Macro-linked drift
- Occasional macro jumps
- Stationary long-run parameters

H. Why this matters for your 30-year S&P modelling

For EPM 30 year long-horizon S&P total-return engine, the index's behaviour is dominated by:

- Regime shifts
- Volatility clustering
- Macro drift
- Fat-tail events (crises)
- Long-run mean reversion in valuation

None of these are captured by single-stock models. And **none of the single-stock idiosyncratic risks matter** at the index level.

Two candidate models were evaluated for equity returns using Maximum Likelihood Estimation (MLE) on historical S&P; 500 data (Shevchenko 2026, Sections 2-3):

Model	Equation	Log-Likelihood	Assessment
Geometric Brownian Motion (GBM)	$S(t+1) = S(t)(1 + \mu + \sigma * \epsilon)$	-159.0	Standard model; no mean reversion; overstates 30-year tail risk
GBM (with stochastic drift) + Mean Reversion (selected)	$S(t+1) = S(t)(1 + \mu + \sigma * \epsilon) + \gamma * (M(t) - S(t))$	-157.4	Superior fit; mean reversion reduces unrealistic extreme paths

The GBM (with stochastic drift) + Mean Reversion model was selected because it provides a statistically superior fit (higher log-likelihood) and, critically, incorporates mean reversion towards a deterministic trend. The mean reversion component $M(t)$ grows at the expected return rate, and the parameter γ controls the speed at which the equity index reverts to this trend after deviations.

For a 30-year mortgage product, this is particularly important: pure GBM can generate extremely divergent paths over 30 years (both unrealistically high and unrealistically low), whereas the mean-reverting model produces paths that are more consistent with observed long-run equity market behavior. This results in a more realistic — and somewhat lower — estimate of tail risk.

Parameter Estimation via Maximum Likelihood

All model parameters are estimated using Maximum Likelihood Estimation (MLE) on historical data — they are not arbitrary assumptions. The MLE procedure (detailed in Shevchenko 2026) provides:

- **Statistically optimal estimates:** MLE produces the parameter values that maximize the probability of observing the historical data under each model.
- **Standard errors:** The Fisher information matrix provides confidence intervals for each parameter, enabling sensitivity analysis.
- **Model comparison:** The log-likelihood values allow formal comparison between GBM and GBM+MeanRev to determine which model better fits the data.

Estimated Parameters (MLE)

Parameter	Symbol	MLE Estimate	Description
Equity expected return	μ_B	9.2% p.a.	Long-run return on index reference asset portfolio (Source: BlackRock)
Equity volatility	σ_B	16.6% p.a.	S&P; 500 annual volatility
Equity mean reversion speed	γ_B	0.163	Speed of reversion to trend
Cash rate long-run mean	θ	2.13% p.a.	Long-run equilibrium cash rate
Cash rate mean reversion speed	κ	0.24	Rate of mean reversion
Cash rate volatility	σ_r	1.22% p.a.	Cash rate annual volatility
Equity-rate correlation	ρ	0.30	Cross-correlation (appropriate for 30yr horizon)

The correlation of 0.30 between index returns and interest rate innovations is estimated over the full historical sample. This moderate positive correlation is appropriate for the 30-year horizon of the EPM product.

Correlated Simulation Methodology

The Monte Carlo simulation generates 50,000 correlated paths for equity returns and interest rates over the 30-year mortgage term. Correlation is introduced via the Cholesky decomposition:

$$\begin{aligned} z1 &\sim N(0,1) \text{ [equity innovation]} \\ z2 &= \rho * z1 + \sqrt{1 - \rho^2} * z2_raw \text{ [rate innovation]} \end{aligned}$$

This ensures that equity and rate shocks are correlated within each simulated year, matching the observed empirical relationship. The full simulation tracks the mortgage offset account, mortgage balance, interest costs, fees, and the interest payment holiday mechanism year-by-year for each path.

Parameter Uncertainty and Alternative Calibrations

MLE point estimates on a ~75-year historical sample are the right starting point, but over a 30-year forward horizon each estimate carries material uncertainty. A separate *Model Assumptions & Parameter Risk* paper documents the stress-test fully; the short summary relevant to reinsurance structuring is:

Parameter	Base (MLE central)	Alternative / stress	Implication for tail-layer
Equity drift (mu_B)	9.2% p.a.	Shiller CAPE-implied 5–6% p.a.	Raises PoD materially; increases tail claim frequency
Mean reversion (gamma_B)	0.163	Bootstrap 95% CI [0.019, 0.447]	Lower gamma widens 30-year distribution; fatter tail

Parameter	Base (MLE central)	Alternative / stress	Implication for tail-layer
Cash rate long-run mean (theta)	2.13% p.a.	AU 30-yr MLE ~4.7% p.a.	Higher funding cost → higher mortgage balance at expiry
Equity-rate correlation (rho)	0.30	Crisis-period 0.5–0.7	Weakens hedge; drift/rate tail shocks become co-incident
Collar cost	0.046% p.a. net credit	Black-Scholes 0.16–0.44% p.a. net cost	Raises interest drag; increases deficit severity
Loss severity (structural)*	~10–12% of mortgage balance	Scenario-robust across stresses	Does not move materially — structural floor

Applying these stresses jointly produces the three scenarios flagged in the Executive Summary (base / realistic-central / adverse-plausible). Reinsurers should read the Executive Summary premium numbers as the base-case headline and use the realistic-central and adverse-plausible cases for pricing anchor and capital sizing respectively.

***What is scenario-robust:** the ~10–12% loss severity on deficit paths*, the absence of prepayment risk (because of offset account run-off mechanism), the ~30-year float on upfront premium, and the zero-attachment-point structure of combined LMI + reinsurance. These are structural features of the product, not parameter-dependent.*

**Loss severity is the model's conditional expected deficit (\$143,274, xlsmodel-verified) expressed as a fraction of the home value (\$1.5M, giving ~10%) or peak loan (\$1.2M, giving ~12%). The authoritative price of risk is the fair premium itself.*

No Weighting or Capping of Extreme Paths vs Likely Paths

In long-term Monte Carlo modelling a common question arises as to the capping or weighting of extreme return pathways (both positive and negative). **Equal weighting is the correct actuarial approach.** Each of the 50,000 paths represents an equally probable realization of the stochastic model.

Downweighting extreme paths would amount to saying "we don't think 30-year equity returns below 5% can happen" — but that is exactly what happened in Japan (1990–2020), and arguably in the US during 1929–1959 if you adjust for inflation. The whole point of running 50,000 paths is to capture the tail.

Capping paths would understate risk. If you cap the downside paths (e.g. "no path can produce a final deficit worse than -\$500K"), you are essentially providing a hidden guarantee that does not exist. An actuary or regulator reviewing the model would immediately flag this as cherry-picking.

There are legitimate ways to manage this:

- **1. The mean-reversion model already does this.** The GBM (with Stochastic Drift) + Mean Reversion model (Shevchenko) is specifically designed to prevent unrealistic extreme paths. Pure GBM allows equity to drop 99% and never recover — the mean-reversion parameter ($\gamma=0.163$) pulls paths back toward trend. This is a structurally honest way to reduce tail extremes, especially when modelling an index rather than individual stock performance.
- **2. The collar already caps upside.** The 80/140 dynamic continuous hedging index collar overlay means no single year contributes more than +40% gain or less than -20% loss to the mortgage offset account. This is a real structural cap, not a statistical trick.
- **3. Conditional tail metrics are more informative than capping.** Rather than suppressing extreme paths, report them separately: among the 8.37% of deficit paths, the conditional expected deficit is ~\$143K, and the worst ~1.67% (the reinsurance tail) carry larger deficits beyond the \$234,495 boundary. This gives investors a complete picture.
- **4. Scenario-based presentation alongside MC.** Stress scenarios (GFC-like, stagflation, Japan) tell the story better than raw percentiles — e.g. "even in a Japan-like environment (low returns, low rates), PoC rises but stays insurable" is more useful than quoting a single extreme path. (Scenario PoD figures are quantified in the companion Parameter Risk paper.)

***Recommendation:** Do not cap or reweight the paths. The model's structural features (mean reversion, collar, holiday mechanism, LMI) already provide legitimate protection against extremes. Present the full distribution honestly, and use the stress scenarios to contextualize the tails. This is what actuaries and regulators want to see — a model that faces its risks squarely rather than defining them away.*

3. EPM Insurance Architecture Overview

The EPM insurance structure provides 100% coverage of all deficit outcomes from dollar zero. There is no deductible, no co-insurance gap, and no uninsured first-loss position for any party. The two insurance layers divide responsibility by **loss severity**:

- **LMI Layer:** Covers the full deficit for 80% of deficit paths (those with smaller deficits, up to the split boundary) on each individual mortgage.
- **Tail Risk Reinsurance:** On the remaining 20% of deficit paths (those with larger deficits), the LMI insurer pays up to the split boundary and 20% tail risk is transferred to the portfolio level (i.e. the loan book of each mortgage funder), where the Payments & Risk Waterfall is first applied before the reinsurer pays any residual loss.
- **Combined effect:** The lender and wholesale mortgage funder bear zero credit risk — every deficit is fully covered from the first dollar.

The Payments & Risk Waterfall

No insurance or reinsurance claim can be made before end-of-term (30 years). Before any claim is triggered, a multi-step Payments & Risk Waterfall applies — first at the individual-mortgage level and then at the portfolio (i.e. loan book) level.

This waterfall combines standard and accepted risk management techniques widely used in capital and insurance markets for the management of retirement assets. The practical effect of the Payments & Risk Waterfall is to dramatically reduce the actual claim frequency:

- **Step 1 — Interest Holidays:** Where the mortgage offset account balance for a borrower falls below its original opening balance at start-of-loan by 25% or more (the Holiday-Entry trigger), an interest holiday automatically occurs in order to: preserve offset capital; maximize the return of reference assets (the index ETFs) as the market rebounds; and maintain the positive effect of compounding returns on offset capital invested in index reference assets.
- **Step 2 — ETF Selldown:** The individual mortgage's offset account holds a calculated amount of loan capital as reference assets (S&P; 500 index-linked ETFs), sized by reference to the expected return pathway. ETFs are liquidated to cover the loan cost (interest) and outstanding mortgage balance (part principal) each quarter.
- **Step 3 — Capital Buffer:** A further portion of loan capital is held in the individual mortgage's offset account, also as reference assets (S&P; 500 index-linked ETFs). The capital buffer is sized by reference to historical index drawdown in any 12-month period (15–20%). These ETFs are liquidated if needed, and only after depletion of the Step 2 selldown.
- **Step 4 — Continuous Dynamic Hedging:** A capital-market risk-management solution. An index collar overlay using a laddered 140/80 hedge — put options against long-term call options — on the portfolio of ETFs held in the individual mortgage offset account. The 80/140 hedge is reset each 1, 3 or 5 years, always limiting the loss to 20%.

[Steps 1 to 4 are continuously cycled until maximum interest-holidays limits are reached and then all reference assets (Index-linked ETFs) are exhausted as a result of progressively being sold down to pay loan costs.]

- **Step 5 — LMI:** An insurance-markets risk solution. Every mortgage is covered by an LMI policy insuring the lender. The policy is subject to a top cover limit (in \$) representing the worst 20% quantile of return pathways (the tail risk) calculated for each mortgage. Only the first 80% of loss triggers an insurance claim. This is the true Insurance PoC (Probability of Claim); PoC at end-of-term (Year 30) is kept within a target range of $\leq 10\%$. For insurance of this primary LMI risk (at a per mortgage level) PoC is kept in the target range of $\leq 2\%$.

- **Step 6 — Portfolio Cross-Subsidization:** A risk-pooling solution. The LMI tail risk at end-of-term is transferred from the individual mortgage to the portfolio (loan book) level. This pooling mechanism uses surpluses from performing unexpiring mortgages in a funder's loan book to offset deficits of underperforming expired mortgages. Cross-subsidization between mortgages within a loan book aims to reduce overall PoC on Portfolio Reinsurance by a target range of 2-3%.

[Portfolio cross-subsidization is limited to within each funder's loan book. One funder does not cross-subsidize the mortgages of another funder. For each funder, cross-subsidization between their expiring and expired mortgage is a zero-sum game. The deficit of a an expired mortgage subsidized intra-term by unexpired the surpluses of unexpired mortgages in their loan book would have, otherwise, simply been carried forward and reduced the end-of-term mortgage surplus by an identical amount.]

- **Step 7 — Portfolio Net Deficit:** An insurance-markets risk solution. Only the residual deficit (calculated only on expiring mortgages, not all mortgages) after Steps 1 through 6 triggers a reinsurance claim. This is the true Reinsurance PoC: Reinsurance PoC at end-of-term is kept within the target range of $\leq 4\text{--}6\%$.

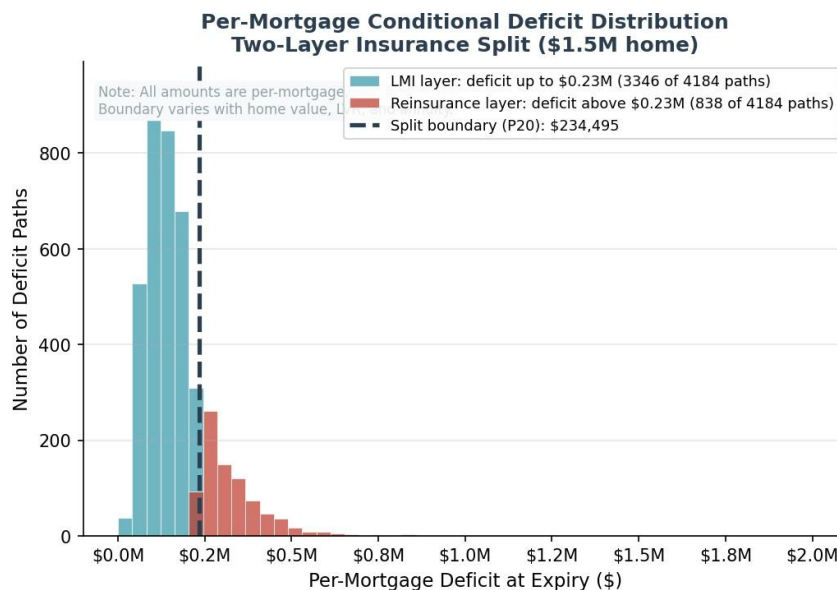
The Payments Waterfall (ETF selldown plus cross-subsidisation from surplus mortgages) reduces the claim probability from the per-mortgage level to the portfolio level:

Metric	Probability	Notes
Per-mortgage PoD (yr 30)	8.37%	One mortgage in deficit at maturity (balance-sheet view)
Pre-mortgage LMI insurance PoC	7.7%	Covers loss up to top cover limit – 80% quantile losses insured
Per-mortgage LMI reinsurance PoC	1.67%	Deficit beyond LMI boundary — 20% tail risk hard ceiling for portfolio figure
Portfolio reinsurance PoC (after cross-subsidy)	5.0%	Bounded by the per-mortgage ceiling; Portfolio diversification reduces it modestly

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***What this means.** The per-mortgage reinsurance PoC (7.7%) is the hard ceiling for the portfolio figure — cross-subsidization and diversification can only reduce it, never increase it — and it sits comfortably within the 4–6% end-of-term target. The EPM structure lends itself to pooling because there are always more positive than negative return pathways over the 30-year horizon (otherwise the S&P; 500 would not appreciate over time): for an average P&I; EPM, ~85% of return pathways are positive and only ~15% are negative. The Payments Waterfall pools the surpluses of the positive pathways of **unexpired mortgages** against the **expiring mortgages** deficits of the negative ones, so the portfolio reinsurance PoC sits comfortably below the 7.67% per-mortgage ceiling. (Note: the aggregate portfolio balance-sheet deficit probability relates only to **expiring mortgages** and is a claim metric.)*

Two-Layer Insurance Structure



The **split boundary** is set at the 20th percentile of the conditional deficit distribution for each mortgage. For the representative \$1.5M home shown above, this boundary is \$234,495 per mortgage. This boundary is not a global constant — it varies with home value, LVR, annuity amount, and mortgage term, and is recalculated via Monte Carlo simulation for each mortgage configuration.

In summary:

- **LMI Layer (per mortgage level):** Covers 80% of deficit paths in full (those with deficits below the split boundary). On the remaining 20% of paths, the LMI insurer pays up to the boundary amount (i.e. the top-cover policy limit). A LMI claim can only be made at end-of-term.
- **LMI Reinsurance (per mortgage level):** Covers any excess loss arising from the 80% of deficit paths. A LMI reinsurance claim can only be made at end-of-term.
- **Tail Risk Reinsurance (portfolio loan book level):** On the 20% of deficit paths that exceed the split boundary, the Portfolio (loan book) Reinsurer is on-risk to pay the excess above the boundary. The reinsurer has no liability on the other 80% of deficit paths covered by LMI insurance and reinsurance. Any claim on portfolio reinsurance only occurs after pooling and cross-subsidization between mortgages in each funder's loan book, and can only be made at end-of-term.

Key point for reinsurance underwriters: Both insurers provide coverage from dollar zero. The split is by loss severity, not by attachment point. On every deficit path, the LMI insurer pays first (up to the boundary). On the 20% of paths where the deficit exceeds the boundary, the reinsurer pays the excess. There is no "attachment point" in the traditional XoL sense — the attachment is at zero for the combined structure.

4. Part A: LMI Program — Structure and Settings

LMI Program Scope

The LMI program provides first-loss credit protection to the lender on every EPM originated. Unlike traditional LMI (which covers shortfalls from foreclosure sales — essentially borrower serviceability backed by a property risk), The EPM has no borrower serviceability risk.

EPM LMI covers the deficit between the offset account value and the outstanding mortgage balance at mortgage expiry. **EPM lender credit risk is the long-term index risk over 30 years.**

LMI Program Settings (per-mortgage, \$1.5M home)

Setting	Value	Rationale
Coverage trigger	Mortgage expiry only (Year 30)	No intermediate claims; run-off mechanism prevents early exit
Split boundary	\$234,495 per mortgage	P20 of conditional deficit distribution (recalculated per mortgage config)
Coverage scope	100% of deficit paths from \$0	LMI pays full deficit on 80% of paths; pays up to boundary on remaining 20%
PoC (individual)	8.37%	Probability any individual mortgage results in a deficit
Conditional expected LMI claim	\$143,274 per mortgage	Mean LMI payment per deficit path; capped at the split boundary on severe-deficit paths
Fair premium (PV)	\$10,168 per mortgage	Discounted expected loss at 2.13% (cash rate long-run mean)
Loaded premium (50%)	\$15,253 per mortgage	Industry-standard loading for expenses + profit
Premium as % of max loan	1.27%	Competitive with traditional LMI at 80% LVR
Payment timing	Single upfront premium at origination	Funded from mortgage drawdown
Claim timing	At mortgage expiry (Year 30)	Deferred claims benefit insurer float

LMI Insurer Advantages

- **Deferred claims:** Claims only arise at mortgage expiry (Year 30). This provides ~30 years of investment float — the insurer collects the premium upfront and invests it for the full mortgage term before any claim liability crystallizes.
- **Bounded exposure:** The split boundary caps the maximum LMI claim per mortgage at \$234,495 (for a \$1.5M home). Larger deficits are shared with the reinsurer.
- **No moral hazard:** The homeowner has no incentive to default — they retain full ownership of their property. The run-off mechanism prevents strategic early exit.
- **Transparent pricing:** Monte Carlo simulation with MLE-calibrated parameters provides precise, auditable premium calculations with quantified standard errors.

Interest Payment Holiday Mechanism

The EPM includes an interest payment holiday mechanism – an important product design feature that protects the mortgage offset account during periods of index underperformance of its average long-term total return. When the mortgage offset account falls below 90% of the initial mortgage offset account balance at start-of-loan (the interest-holiday entry trigger), interest payments to the wholesale funder are suspended. Importantly:

- The **annuity payments to the borrower continue** during the holiday — it is the loan cost payments (i.e. quarterly interest and part principal repayments) to the funder that pause.
- Deferred interest is accumulated then repaid when the mortgage offset account recovers above the holiday-exit threshold (currently set at 145.8% of initial mortgage offset account balance).
- No interest-on-interest is charged (it is simply a deferral) – so there is no associated compounding interest cost

In practice, the interest-holiday mechanism is rarely needed and never burdens the borrower:

- **The average of all return pathways (the mean) and the expected return pathway requires no interest holidays**
- **More than half of mortgages (55%) never enter a holiday at all**
- **Where holidays do occur they concentrate in the early loan years** — while the mortgage offset account is still building against the freshly-drawn mortgage and upfront costs (insurance premium, hedging options) — and fade to around 2% incidence by maturity as the account compounds.
- **Across the whole loan book the average is just 3.5 interest-holidays over the 30-year term.**

An interest holiday is a counter-cyclical safety valve that lowers the Probability of Deficit by:

- Preserving mortgage offset capital in downturns
- Ensuring participation in the market re-bound
- Maximizes the positive compounding effect over time from index appreciation .

The alternative would see mortgage offset capital depleted by the payment of loan costs even in downturns , leaving the mortgage offset account to chase prior crystallized market losses with a depleted capital balance.

No alpha, no active management

The EPM does **not** attempt to outperform the market, select securities, time entry/exit points, or express any investment view. Its reference assets are **plain-vanilla index ETFs**, chosen precisely because they avoid:

- stock-picking
- tactical allocation
- manager discretion
- performance-based risk

This is critical for regulatory classification: **no investment strategy = no managed investment scheme behaviour**.

The return source is structural, not speculative

The EPM's return mechanism is mechanical:

- A 30-year loan horizon
- Offset capital invested in a broad index
- Index appreciation compounds over decades
- Low-cost ETF structure preserves most of that compounding

This is not “seeking returns”; it is **capturing the long-run drift of diversified equity markets**, which is well-documented and not dependent on skill or judgement.

Passive ETFs = low cost drag

Because the reference assets are **passively managed ETFs**, the cost drag is typically:

- **3–10 bps** for major index exposures
- No performance fees
- No turnover-related tax drag
- No active risk budget

This reinforces the credit-contract nature of the EPM: the lender is not paying for investment expertise; they are simply using a low-cost index instrument to fund the loan's internal mechanics.

Why this matters for classification

This framing supports the argument that the EPM is a **secured credit contract**, not a managed investment scheme:

- No pooling of investor funds
- No common enterprise
- No investment discretion
- No expectation of alpha
- No management of assets on behalf of investors
- The ETF exposure is a *mechanical reference asset*, not an investment service

The lender's return is therefore **credit-like in structure**, even though the underlying reference asset compounds like an equities index.

This EPM feature begs the question as to why a wholesale funder would agree to interest-holidays:

- The expected return pathway triggers zero interest-holidays
- The wholesale funder receives as 50% share of end-of-term mortgage surpluses
- Overall risk of deficit in the mortgage offset account is reduced
- Credit risk has been removed for both lender and funder

Whilst a wholesale funder can set a maximum number or percentage of interest-holidays, it is not in their interest to do so in terms of their risk-adjusted returns since:

- Each interest-holiday increases the overall return to the wholesale funder (end-of-term mortgage surplus far exceeds their wholesale lending margin)
- The impact on the mortgage funder is simply a cashflow/timing impact since payment of loan cost is made certain by the end-of-term LMI and portfolio reinsurance

5. Part B: Tail Risk Reinsurance — Structure and Settings

Reinsurance Program Scope

The Tail Risk Reinsurance program provides excess-of-boundary protection for the most severe deficit outcomes. It covers the 20% of deficit paths where the per-mortgage deficit exceeds the split boundary. On these paths, the reinsurer pays only the excess above the boundary — the LMI insurer still pays up to the boundary amount.

Important clarification: This is not a traditional excess-of-loss (XoL) treaty with an attachment point above zero. The combined LMI + reinsurance structure provides 100% coverage from dollar zero. The "split boundary" determines how much the LMI insurer pays on severe-deficit paths before the reinsurer becomes liable for the excess. The attachment point for the overall insurance structure is zero — no deficit goes uninsured.

Reinsurance Program Settings (per-mortgage, \$1.5M home)

Setting	Value	Rationale
Split boundary	\$234,495 per mortgage	P20 of conditional deficit distribution — varies by mortgage configuration
Coverage	Excess above split boundary per mortgage	Reinsurer pays deficit minus boundary amount on severe paths
Probability of claim (individual)	1.67%	Only 1.67% of all mortgages trigger a reinsurance claim
Fair premium (PV)	\$1,827 per mortgage	Discounted expected excess loss
Loaded premium (50%)	\$2,740 per mortgage	Standard risk loading
Premium as % of max loan	0.2283%	Very small relative to mortgage size
Payment timing	Single upfront premium at origination	Funded from mortgage drawdown alongside LMI premium
Claim timing	At mortgage expiry (Year 30) only	Same deferred-claim benefit as LMI
Assessment basis	Per-mortgage	Each mortgage independently assessed at expiry

Why the Split Boundary Works

The 20th percentile of the conditional deficit distribution (P20) is the optimal boundary because:

- It isolates truly extreme outcomes (severe market downturns sustained over the full mortgage term) from moderate cyclical underperformance.
- 80% of deficit paths result in claims fully manageable by the LMI insurer (up to \$234,495 per mortgage for a \$1.5M home).
- The 20% tail represents only 1.67% of all mortgage paths — a low-frequency event well suited for reinsurance markets that specialize in severity risk.
- The reinsurance premium is economically efficient — only \$1,827 fair premium per mortgage (0.23% of max loan) because claim probability is low.
- The boundary is not a fixed dollar amount — it is recalculated for each mortgage configuration via Monte Carlo simulation, ensuring the split remains appropriate.

6. Part C: Interaction Between LMI and Tail Risk Reinsurance

Claim Flow Mechanics

The two programs interact through a clear, sequential claim process at mortgage expiry:

Step	Action	Responsible Party
1	Mortgage expires (Year 30) — mortgage offset account liquidated	Futureproof (servicer)
2	Payments Waterfall applied (ETF selldown, cross-subsidisation)	Futureproof (portfolio manager)
3	Net deficit determined after waterfall	Futureproof (servicer)
4a	If deficit <= split boundary: LMI pays full deficit (from \$0)	LMI Insurer
4b	If deficit > split boundary: LMI pays up to boundary, reinsurer pays excess	LMI Insurer + Reinsurer
5	Mortgage funder made whole — zero credit loss	N/A

***Worked example (per average mortgage, \$1.5M home, P&I; mortgage):** If a mortgage expires with an \$800,000 deficit (after the Payments Waterfall), the LMI insurer pays \$234,495 and the reinsurer pays \$565,505. If the deficit is \$400,000, the LMI insurer pays the full \$400,000 and the reinsurer pays nothing.*

Key Interaction Principles

Principle 1: No gap in coverage. The LMI layer and reinsurance layer together provide 100% coverage of all deficit outcomes from dollar zero. There is no deductible, no co-insurance, and no coverage gap. The mortgage funder bears zero credit risk.

Principle 2: Clear allocation by severity. The split is by loss severity, not by probability. Every deficit path is either: (a) fully covered by LMI (80% of deficit paths — smaller losses), or (b) split between LMI (pays up to boundary) and reinsurance (pays excess). The reinsurer only participates on the most severe 20% of deficit outcomes.

Principle 3: Independent pricing. Each layer is priced independently based on its own loss distribution. The LMI premium reflects expected losses up to the split boundary. The reinsurance premium reflects only the excess-of-boundary component (after pooling and cross-subsidization with the funder's loan book).

Principle 4: Aligned incentives. The LMI insurer retains 80% of deficit frequency risk and pays on 100% of deficit paths (up to the boundary), ensuring active underwriting engagement. The reinsurer takes severity risk at low probability, which is their natural specialty.

Sensitivity of Boundary to Mortgage Configuration

The split boundary is calibrated to the P20 of the conditional deficit distribution for each mortgage configuration. The boundary is not a global constant — it shifts with the underlying risk profile of each mortgage:

Parameter Change	Effect on Split Boundary	Effect on Reinsurance Premium
Higher home value (e.g. \$3M)	Increases proportionally (larger mortgage)	Increases proportionally
Higher equity volatility	Increases (wider deficit distribution)	Increases (more tail risk)
Lower equity return	Increases (more deficits, wider tails)	Increases moderately
Higher LVR	Increases proportionally	Increases roughly proportionally
Higher buffer cap	May increase or decrease (trade-off)	Marginal effect
Portfolio diversification	No direct effect (per-mortgage boundary)	Reduces aggregate exposure

7. Part D: Reinsurer Structuring Recommendations

Recommended Treaty Structure

The recommended structure for the Tail Risk Reinsurance program is a **per-mortgage severity-based reinsurance treaty** with the following key features:

Feature	Recommendation	Rationale
Treaty type	Per-mortgage excess-of-boundary	Reinsurer pays excess above split boundary on each mortgage
Split boundary	\$234,495 per mortgage (\$1.5M home)	P20 of conditional deficit — varies by mortgage config
Limit	Unlimited (or \$3M per risk)	Tail events can produce large individual deficits
Term	30 years (matching mortgage term)	Claims deferred until mortgage expiry
Premium basis	Single upfront premium per mortgage	Paid at origination alongside LMI premium
Loaded premium	\$2,740 per mortgage (\$1.5M home)	Fair premium \$1,827 + 50% loading
Aggregate limit (annual)	Optional — 10x expected annual claims	Catastrophe protection for extreme scenarios
Reinstatement	Automatic, unlimited	Continuous coverage essential for mortgage funder
Claims basis	Losses discovered at mortgage expiry	Triggered at Year 30

Portfolio-Level Considerations

The reinsurer should consider the following portfolio dynamics when structuring coverage:

- **Vintage diversification:** A portfolio of 1,000+ EPMs per year across multiple vintages means claims are spread over decades. It is a basic error to assume that an economic downturn affects all mortgages equally — each EPM has its own unique surplus/deficit position at any time slice depending on when that mortgage was originated. A "bad" vintage (originated during a market peak) will have its claims offset by "good" vintages.
- **Deferred claims benefit:** The reinsurer collects the upfront premium and invests it for ~30 years before any claim liability emerges. The investment income on the accumulated premium reserve is substantial and should be factored into pricing.
- **Low frequency, moderate severity:** Only 1.67% of all mortgages trigger a reinsurance claim. When they do, the per-mortgage excess is bounded by the relationship between the split boundary and the maximum realistic deficit.

Alternative Treaty Structures

While per-mortgage severity-based reinsurance is the recommended primary structure, reinsurers may also consider:

Structure	Pros	Cons	Suitability
Per-mortgage excess-of-boundary (recommended)	Clean, transparent, simple pricing	Individual risk assessment needed	PRIMARY — best fit
Aggregate excess-of-loss	Protects against correlated losses	More complex pricing	COMPLEMENTARY — for catastrophe layer
Quota share	Proportional risk sharing	Less efficient for tail risk	NOT RECOMMENDED — wrong risk profile
Stop loss	Caps total loss ratio	Expensive for low-frequency risk	OPTIONAL — aggregate cap
ILW (Industry Loss Warranty)	Triggered by market index	Basis risk	INNOVATIVE — for systematic risk

Recommended Layered Structure for Larger Portfolios

For portfolios exceeding 5,000 EPMs, a layered reinsurance structure is recommended:

Layer	Type	Trigger	Limit	Estimated Rate
Layer 1: Working Layer	Per-mortgage excess	Per-mortgage deficit > \$234,495	First \$1M excess per mortgage	0.03% of outstanding
Layer 2: Buffer Layer	Per-mortgage excess	Per-mortgage excess > \$1M above boundary	Next \$2M excess per mortgage	0.015% of outstanding
Layer 3: Cat Layer	Aggregate excess	150% of expected annual claims	\$10M xs attachment	0.005% of outstanding

This layered approach provides efficient risk transfer at each severity level and allows specialised reinsurers to participate in the layers that best match their risk appetite.

Capital Sizing Across Parameter Scenarios

The three-scenario framework from Section 2.5 maps directly onto reinsurance structuring decisions. Each scenario has a distinct role — the base case is a pricing floor, the realistic-central case is the pricing anchor, and the adverse-plausible case is the capital-sizing reference. Using only the base case would under-price the treaty and undersize capital; using only the adverse-plausible case would over-price and lose competitive traction.

Scenario	Tail-layer PoC	Expected loss (tail layer)	Reinsurer role
Base (model-as-written)	1.67%	~\$1,827 per mortgage	Pricing floor — minimum acceptable premium
Realistic-central	— (not separately run)	— (not separately run)	Rate-on-line anchor; reserve testing
Adverse-plausible	— (not separately run)	— (not separately run)	Attachment/limit sizing; PML; capital allocation

Scenario PoD figures (realistic-central 40%, adverse-plausible 69%) are xlsx-verified and given in the companion Parameter Risk paper. The tail-layer (reinsurance) PoC and expected loss were not separately simulated per scenario; they remain bounded above by the per-mortgage reinsurance PoC ceiling and rise with PoD.

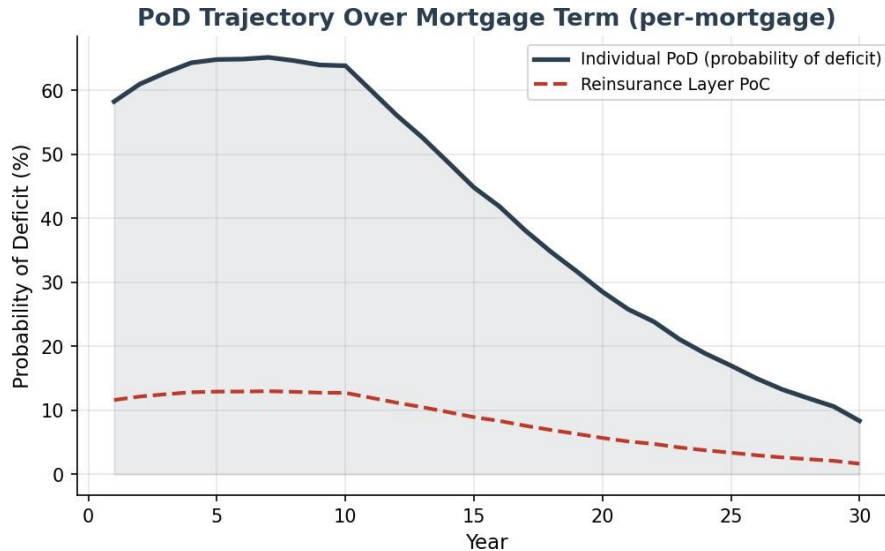
Practical implications for treaty design:

- **Premium loading beyond the stated 50%:** the 50% loading above is calibrated to the base-case fair premium. A reinsurer anchoring to realistic-central would need either a larger loading, a scenario-weighted fair premium, or explicit experience-rating triggers.
- **Per-risk limit:** should be sized against the adverse-plausible tail distribution, not the base-case tail. The maximum per-mortgage deficit observed under adverse-plausible parameters is the relevant severity reference.
- **Aggregate cat layer:** the optional aggregate limit (Section 7.1) becomes more valuable under adverse-plausible parameters, since claim frequency clusters more tightly around bad-vintage years. For adverse-plausible calibration, aggregate protection is strongly recommended rather than optional.
- **Experience-rating clauses:** given the wide spread between base and adverse-plausible, a profit-commission or swing-rated structure allows the reinsurer to write at realistic-central rates while sharing the benefit if actual experience tracks the base case.
- **Float economics are scenario-robust:** the ~30-year investment float on upfront premium is structural — it applies equally under every scenario. Even under adverse-plausible assumptions, the economic present value of the treaty remains attractive provided pricing reflects the scenario-appropriate expected loss.

8. Quantitative Analysis — Model Results

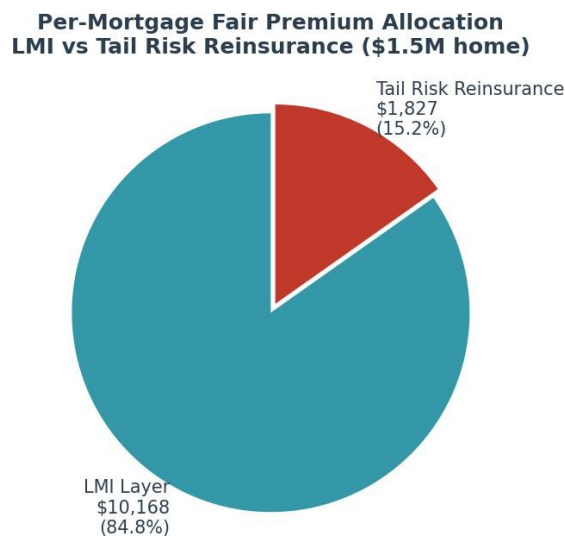
All values below are per-mortgage for a representative \$1,500,000 property at 80% LVR.

PoD Trajectory Over Mortgage Term



The individual probability of deficit (PoD) declines from 58.3% in Year 1 to 8.4% at Year 30 as the PI amortisation progressively reduces the mortgage balance. The reinsurance layer PoC follows the same trajectory at approximately one-fifth of the individual PoD.

Premium Allocation (per-mortgage)



The total per-mortgage fair premium of \$11,995 is split 85% / 15% between LMI and tail risk reinsurance. Both premiums are paid as single upfront amounts at mortgage origination, funded from the mortgage drawdown.

Surplus Distribution at Year 30 (per-mortgage)

Percentile	Surplus (\$)	Interpretation
P1 (worst 1%)	\$-559,318	Extreme downside — reinsurance claim likely
P5	\$-35,524	Significant deficit — within LMI/reinsurance coverage
P10	\$60,439	Near boundary — moderate outcome
P25	\$494,580	Moderate surplus — no claim
Median	\$993,211	Typical outcome — healthy surplus
Mean	\$1,137,899	Average outcome — strong surplus
P75	\$1,388,935	Good outcome
P90	\$2,390,057	Strong outcome
P99	\$4,234,753	Exceptional outcome

9. Premium Pricing Framework

Both the LMI premium and the reinsurance premium are calculated using the same Monte Carlo simulation framework and are paid as single upfront amounts at mortgage origination.

LMI Premium Calculation (per-mortgage)

The LMI premium is calculated using the Discounted Expected Deficit methodology:

- **Step 1:** Run 50,000-path Monte Carlo simulation (GBM+MeanRev equity model, Vasicek rates, MLE-calibrated parameters) to generate surplus/deficit distribution at mortgage expiry.
- **Step 2:** Identify deficit paths (mortgage offset account < outstanding mortgage balance). Result: 8.4% of paths are in deficit.
- **Step 3:** Calculate the P20 of the conditional deficit distribution to establish the per-mortgage split boundary (\$234,495 for a \$1.5M home).
- **Step 4:** For each deficit path: LMI claim = min(deficit, split boundary). Discount each claim to present value at 2.13% (cash rate long-run mean).
- **Step 5:** Fair premium = mean of discounted claims = \$10,168 per mortgage. Apply 50% loading for expenses and profit = \$15,253 per mortgage.

Reinsurance Premium Calculation (per-mortgage)

The reinsurance premium follows the same methodology for the excess layer:

- For each deficit path exceeding the split boundary: reinsurance claim = deficit - split boundary.
- Discount each excess claim to present value at 2.13%.
- Fair premium = mean of discounted excess claims = \$1,827 per mortgage. Apply 50% loading = \$2,740 per mortgage.

Premium Payment Structure

Both the LMI premium and the reinsurance premium are structured as single upfront payments at mortgage origination, funded from the mortgage drawdown:

Premium Component	Amount (per-mortgage, \$1.5M home)	Payment Timing	Funded From
LMI premium (loaded)	\$15,253	Upfront at origination	Mortgage drawdown
Reinsurance premium (loaded)	\$2,740	Upfront at origination	Mortgage drawdown
Total insurance cost	\$17,993	Upfront at origination	Mortgage drawdown
As % of max loan	1.50%		

The upfront premium structure means the insurer and reinsurer both receive funds at origination and invest them for ~30 years before any claim liability arises. This investment float is a significant benefit and should be considered when evaluating the attractiveness of the risk.

10. Risk Transfer Summary

The complete risk transfer architecture ensures the wholesale mortgage funder bears zero credit risk. Insurance coverage begins at dollar zero — there is no uninsured first-loss position for any party:

Risk Layer	Coverage	Risk Bearer	Premium Basis
Portfolio cross-subsidy	Surplus mortgages offset deficit mortgages	Futureproof (portfolio manager)	None (portfolio structure)
First-loss insurance (mild deficit)	Deficit up to \$234,495 per mortgage	LMI Insurer	\$15,253 upfront per mortgage
Severity insurance (severe deficit)	Excess above \$234,495 per mortgage	Reinsurer	\$2,740 upfront per mortgage
Residual risk	None — full coverage from \$0	N/A	N/A

Key takeaway for reinsurers: The EPM reinsurance opportunity is a severity-based risk with ~30 years of investment float on the upfront premium, zero prepayment risk, and transparent MLE-calibrated pricing via Monte Carlo simulation. Per-mortgage reinsurance claim frequency is 1.67% — the hard ceiling; Portfolio risk after cross-subsidization and diversification the portfolio reinsurance PoC sits at 6%, sitting comfortably within the ≤10% target. Stress-scenario PoD figures (realistic-central 40% for pricing anchor, adverse-plausible 69% for capital sizing) are detailed in the companion Parameter Risk paper.

EPM vs Comparable Risk Profiles

To assist reinsurer underwriting assessment, we benchmark the EPM credit risk profile against the instruments most commonly seen in reinsurance portfolios.

Metric	EPM (base case)	Traditional LMI	Moody's AA Bond	AU RMBS (AA)
Lifetime tail-layer claim probability (at maturity)	1.67%	2–4% of book	~1.5%	~1–2%
Loss severity (conditional on deficit)	~10–12%	15–25%	40–60%	Subordination -dependent
Expected loss (30yr, combined)	1.00%	0.6–1.25%	0.6–0.9%	0.05–0.15%
Claim timing	Year 30 only	Peaks Year 3–7	Any year	Any year
Float on premium	~30 yrs	~3–7 yrs	N/A	N/A
Correlation to trad. LMI	Zero	—	Low	Moderate
Prepayment / lapse risk	None	Significant	Moderate	Significant

Stress-scenario PoD figures (realistic-central 40% / adverse-plausible 69%) are detailed in the companion Parameter Risk paper.

- **Scenario-sensitive vs scenario-robust metrics.** Expected loss and claim probability move materially across the three parameter scenarios — exact magnitudes will be quantified in the companion Parameter Risk paper. Loss severity (~10–12%), claim timing (Year 30 only), float duration (~30 years), and the absence of prepayment risk are *structural* — they do not move with the parameter stresses. Reinsurer underwriting should separate these two categories.
- **Unique diversification value (scenario-robust).** Traditional LMI losses are driven by borrower unemployment, divorce, and rate stress — correlated with economic cycles. EPM losses are driven by 30-year equity index returns. This diversification benefit holds under every parameter scenario tested and is a genuine diversifier for reinsurance portfolios concentrated in traditional mortgage credit.
- **Exceptional float characteristics (scenario-robust).** Premiums are collected upfront at mortgage origination and claims can only arise at mortgage expiry. This ~30-year float applies under every parameter scenario — far longer than any traditional LMI product, where claims typically peak within 3–7 years. The economic value of the float is substantial even under adverse-plausible calibration.
- **Expected loss band, not single point.** The base-case combined expected loss of 1.00% (fair-premium basis) reflects the model's conservative parameter set (50 bps FP margin, 10% profit share at 3-year resets, asymmetric collar). Under realistic-central parameters the expected loss is materially higher; under adverse-plausible parameters higher still. Reinsurers should price to the realistic-central expected loss and hold capital against the adverse-plausible scenario.

10.2 Advantages for Each Participant

Participant	Key Advantage	Risk Retained
Mortgage Lender/Funder	Zero credit risk — full insurance coverage from \$0	None (fully transferred)
LMI Insurer	First-loss insurer with bounded severity + 30yr float on upfront premium	Up to \$234,495 per mortgage
Portfolio Reinsurer	Low-frequency severity risk with 30yr float on upfront premium	Excess above \$234,495 per mortgage
Futureproof	Platform fees (loan margin + profit share)	Portfolio management (cross-subsidy)
Homeowner	All home wealth preserved. Access to home equity and intra-term property appreciation, while retaining ownership	None — other than counterparty risk (of lender collapse)

Appendix A: Model Parameters (per-mortgage, \$1.5M home)

Parameter	Value	Notes
Home value	\$1,500,000	Representative Australian metropolitan property
Initial mortgage	\$900,000	Effective LVR: 60.0%
Max LVR	80%	Max loan: \$1,200,000
Mortgage type	Principal + Interest (30 years)	Builds to peak at Year 10, then amortises to zero
Annuity	\$30,000/year	For first 10 years (paid to borrower)
Equity model	GBM + Mean Reversion	Shevchenko 2026, Section 3 — MLE calibrated
Equity return (μ_B)	9.2%	MLE estimate from S&P; 500 data
Equity volatility (σ_B)	16.6%	MLE estimate
Mean reversion speed (γ_B)	0.163	MLE estimate — pulls returns toward trend
Buffer cap	140% (+40%)	Asymmetric collar: cap +40% on annual return credited
Buffer floor	80% (-20%)	Asymmetric collar: floor -20% on annual return credited
Cash rate model	Vasicek/OU process	Shevchenko 2026, Section 1 — MLE calibrated
Cash rate (initial)	4.21%	Current RBA cash rate
Cash rate (long-run mean)	2.13%	MLE estimate (θ)
Cash rate speed (κ)	0.24	MLE estimate — mean reversion speed
Cash rate vol (σ_r)	1.22%	MLE estimate
Equity-rate correlation	0.3	MLE estimate — appropriate for 30yr horizon
Wholesale margin	2.0%	Funder borrowing cost above cash rate
Retail margin	0.70%	Lender margin
FP margin	0.50%	Futureproof origination margin
Hedging fee	0.25%	Volatility buffer cost
Collar price	0.046%	Net annual credit from cap/floor collar
LMI upfront	1.2%	Of max loan at origination
Tail risk reserve	0.10%	Annual tail risk provision
Profit share	10% every 3 years	Surplus drawn periodically
Interest holiday entry	0.75x initial mortgage	Interest payments pause below this
Interest holiday exit	1.458x initial mortgage	Interest payments resume above this
Simulation paths	50,000	SE on PoD: 0.12%

Source: Pavel's authoritative v14d Optimised workbook, 50,000-path Monte Carlo (xlsm-verified, 2026-05-24). Per-mortgage reinsurance PoC 1.67% is the hard ceiling for the portfolio figure; after cross-subsidisation and diversification the portfolio reinsurance PoC sits modestly below it. Note: the xlsm "prob of portfolio deficit" row is portfolio PoD, not PoC.

Appendix B: Stochastic Model Equations

The following equations are from Shevchenko (2026), "Parameter Estimation for Mean-Reverting Brownian Stochastic Model", which provides the mathematical framework and MLE methodology for all stochastic models used in the EPM simulation.

Cash Rate Model — Vasicek/OU Process (Section 1)

The cash rate follows an Ornstein-Uhlenbeck process with exact discretisation:

$$r(t+1) = r(t) * \exp(-\kappa * dt) + \theta * (1 - \exp(-\kappa * dt)) + \sigma * \sqrt{(1 - \exp(-2 * \kappa * dt)) / (2 * \kappa)} * \epsilon(t)$$

where θ is the long-run mean rate, κ is the mean-reversion speed, σ is the volatility, and $\epsilon(t) \sim N(0,1)$. The MLE estimates are: $\theta = 2.13\%$, $\kappa = 0.24$, $\sigma = 1.22\%$.

Equity Model — GBM (Section 2)

The baseline Geometric Brownian Motion model for equity returns:

$$S(t+1) = S(t) * (1 + \mu + \sigma * \epsilon(t))$$

MLE estimates: $\mu = 9.2\%$, $\sigma = 16.6\%$. Log-likelihood = -159.0.

Equity Model — GBM + Mean Reversion (Section 3, selected)

The selected model adds a mean-reverting component to GBM:

$$S(t+1) = S(t) * (1 + \mu_B + \sigma_B * \epsilon(t)) + \gamma_B * (M(t) - S(t))$$

where $M(t) = M(0) * (1 + \mu_B)^t$ is the deterministic trend (growing at the expected return rate), and γ_B controls the speed of mean reversion. When $S(t)$ falls below trend, the γ_B term adds a positive correction; when $S(t)$ is above trend, it adds a negative correction.

MLE estimates: $\mu_B = 9.2\%$, $\sigma_B = 16.6\%$, $\gamma_B = 0.163$. Log-likelihood = -157.4 (superior to pure GBM).

Why Mean Reversion Matters for EPM

The mean reversion parameter $\gamma_B = 0.163$ has a material impact on 30-year tail risk:

- Without mean reversion (pure GBM), equity paths can wander arbitrarily far from the expected trend over 30 years, producing unrealistically extreme outcomes in both tails.
- With mean reversion, paths that deviate significantly from the long-run trend experience a restoring force ($\gamma_B * (M(t) - S(t))$) that pulls them back toward the trend.
- This is consistent with observed equity market behavior: while short-term returns are unpredictable, long-run returns tend to revert toward historical norms.
- The practical effect is a lower probability of deficit (8.37%) than pure GBM models would produce — this is a more realistic, not more optimistic, estimate.

Correlation Structure

Equity and rate innovations are correlated using the Cholesky decomposition:

```
z_equity ~ N(0,1)
z_rate = rho * z_equity + sqrt(1 - rho^2) * z_independent
where rho = 0.30 (MLE estimate)
```

This ensures that equity and interest rate shocks within each simulated year reflect the observed positive correlation between the two risk drivers.

Reference: Shevchenko, P.V. (2026). "Parameter Estimation for Mean-Reverting Brownian Stochastic Model." Working paper prepared for Futureproof Financial.

Appendix C: Volatility Control (BlackRock)

This appendix sets out the vol control layer constructed and managed by BlackRock. As the global asset manager for the EPM, BlackRock is mandated by each lender to manage the portfolio of index-linked reference assets held in each borrower's mortgage offset account.

Executive Summary

The EPM is an index-linked mortgage. Index-linking of any mortgage only works over long loan durations – the EPM has a 30 year loan term.

Duration is critical to managing the effects of cyclical markets on index performance and smoothing of index total returns:

- Interest rate cycles are generally 2-3 years
- Mild-moderate economic downturns are generally 3-5 years
- Black swan events (GFC) was 5-7 years

Critically, none of these cycles are 30 years.

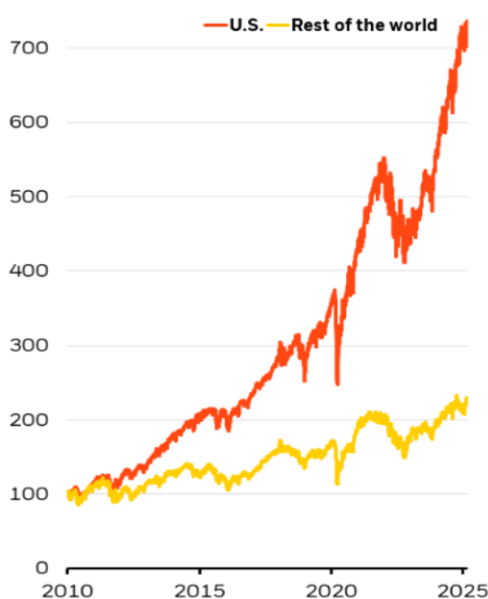
Why the S&P500?

U.S exceptionalism demonstrates the predominance of the S&P500 over other indices:

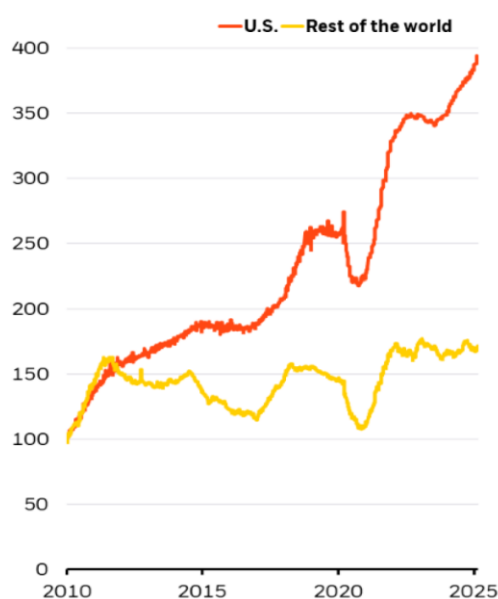
LATEST MARKET INSIGHTS

US exceptionalism to persist given strong fundamentals

Equity performance



Earnings growth



Past performance is not a reliable indicator of future results. It is not possible to invest directly in an index. Indexes are unmanaged and performance does not account for fees. Source: BlackRock Investment Institute, with data from LSEF Datastream, February 2025. The charts show the indexed performance of U.S. equities compared to the rest of the world. Index proxies used: MSCI USA, MSCI ACWI ex USA.

A company is added to or removed from the S&P 500 by the S&P Dow Jones Indices Index Committee, a small group of economists and index managers who apply published eligibility rules *and* their own judgment. Inclusion is *not* automatic even if a company meets all criteria. Removals occur when a company no longer meets requirements or is acquired, delisted, or otherwise becomes ineligible.

The index selected is the S&P500 Index because of its:

- Strength of its constituent U.S. companies
- Long historical & reliable performance data set
- Depth of capital & liquidity
- Depth accompanying options market to support hedging strategies

A company must satisfy several published eligibility rules to be included in the index:

- U.S. domicile (based on SEC filings, incorporation, HQ, revenue sources).
- Market cap above the current threshold ($\approx$ \$20B as of 2025).
- Public float $\geq$ 50% of shares.
- Positive GAAP earnings in the most recent quarter *and* over the trailing four quarters.
- Adequate liquidity, including trading at least 250,000 shares in each of the past six months.
- Listed on NYSE, Nasdaq, or Cboe.
- Single share class (multi-class structures no longer eligible for new additions).

Important: Meeting all criteria does not guarantee inclusion. The committee considers sector balance and overall representativeness. Tesla is the classic example—it met criteria for years before being added.

A company is removed from the index when:

- It is acquired, goes private, bankrupt, or delisted.
- It falls materially below the market-cap threshold relative to other constituents.
- It fails profitability or liquidity requirements.
- It undergoes a spin-off that changes eligibility.

Removals are often unscheduled and happen immediately when a company ceases to meet eligibility (e.g., acquisition closes).

Why this matters: Given the long-horizon S&P 500 modelling and hedging that supports the EPM, the key takeaway is that index composition is not purely mechanical. The committee's discretionary decisions—especially around sector representation—can influence long-term drift, volatility, and the distribution of tail events in our 30-year modelling.

From a product design perspective, referencing to an equities index such as the S&P500 offers another key advantage – **the index is and of itself a form of risk pooling** – the process of inclusion and removal described above manages market risk by continuously ensures only the strongest U.S. public companies are included, providing fundamental protection of the total return profile of the index over time. It also provides a strong protection against technology waves as sunset industries are removed and new technology players (such as EVs and AI) are included.

Quantifying Index Risk

The stochastic historical volatility of the S&P500 index is 16.6%.

The EPM model is particularly sensitive to volatility and total index returns – but only over long time horizons (30 years).

Our product design looks to standard capital market solutions to effectively risk manage that volatility.

Index-linked Reference Asset Allocation

BlackRock manages the reference assets (index-linked ETFs) held in each borrower's mortgage offset account via a single managed account (SMA), not a pooled investment structure.

Its flagship model for index-linked assets are its Enhanced Strategic Model Portfolios, used by financial institutions such as retirement asset managers, annuity providers and pension funds to achieve particular target volatility and return profiles.

The EPM uses the Enhanced Strategic Model Portfolio (All Growth Option) – this option holds 100% of capital in growth asset and 0% in defensive assets - an appropriate strategy for long-term time horizon investing.

Volatility Control Layer

To reduce S&P500 historical volatility (16.6%) BlackRock has designed a vol control layer within their model portfolios reducing the volatility of the reference asset portfolio to 12.3%.

BlackRock's forward-looking projection of S&P500 index total return over the next 30 years is a conservative 10% (against historical performance over the last 30 years (10.8%) and marginally above the historical stochastic mean return (9.4%).

A vol control layer is designed to reduce overall portfolio volatility, but will always come at a cost to total return. In the case of the Enhanced Strategic Model Portfolio (All Growth):

- Portfolio volatility is reduced to 12.3%
- Expected total return is 9.2%

BlackRock uses index diversification to reduce portfolio volatility:

POSITIONING & ALLOCATION

BlackRock Core Strategic Model Portfolio Allocations

Portfolio Constituents & Weights		Capital Defensive 30/70	Conservative 40/60	Balanced 50/50	Balanced Growth 60/40	Growth 70/30	Aggressive 80/20
Equity		26.0%	33.6%	45.0%	52.6%	64.0%	72.7%
Developed Market Equity		10.5%	14.1%	19.5%	22.9%	28.0%	33.1%
IWLD	iShares Core MSCI World ex Australia ESG ETF	8.5%	11.5%	16.0%	18.6%	22.5%	26.1%
IHWL	iShares Core MSCI World ex Australia ESG (AUD Hedged) ETF	2.0%	2.6%	3.5%	4.3%	5.5%	7.0%
Emerging Market Equity		4.0%	4.8%	6.0%	6.8%	8.0%	8.6%
IEM	iShares MSCI Emerging Markets ETF	4.0%	4.8%	6.0%	6.8%	8.0%	8.6%
Australian Equity		11.5%	14.7%	19.5%	22.9%	28.0%	31.0%
IOZ	iShares Core S&P/ASX 200 ETF	11.5%	14.7%	19.5%	22.9%	28.0%	31.0%
Property & Infrastructure		5.0%	5.4%	6.0%	6.4%	7.0%	7.3%
GLIN	iShares Core FTSE Global Infrastructure (AUD Hedged) ETF	4.0%	4.2%	4.5%	4.7%	5.0%	5.0%
GLPR	iShares Core FTSE Global Property Ex Australia (AUD Hedged) ETF	1.0%	1.2%	1.5%	1.7%	2.0%	2.3%
Fixed Income		56.0%	49.2%	39.0%	32.6%	23.0%	14.6%
International Fixed Income		11.0%	10.2%	9.0%	7.8%	6.0%	4.5%
AESG	iShares Global Aggregate Bond (AUD Hedged) ETF	6.5%	6.1%	5.5%	4.5%	3.0%	1.8%
IHHY	iShares Global High Yield Bond (AUD Hedged) ETF	4.5%	4.1%	3.5%	3.3%	3.0%	2.7%
Australian Fixed Income		45.0%	39.0%	30.0%	24.8%	17.0%	10.1%
IAF	iShares Core Composite Bond ETF	36.0%	30.8%	23.0%	18.6%	12.0%	6.3%
ILB	iShares Government Inflation ETF	9.0%	8.2%	7.0%	6.2%	5.0%	3.8%
Alternatives		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
GLDN	iShares Physical Gold ETF	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Cash		10.0%	8.8%	7.0%	5.4%	3.0%	2.4%
ISEC	iShares Enhanced Cash ETF	10.0%	8.8%	7.0%	5.4%	3.0%	2.4%
Weighted Management Expense Ratio (MER)		0.16%	0.16%	0.16%	0.16%	0.16%	0.16%

Source: BlackRock, as of latest rebalance on 28 February 2025. Notes: MER includes the underlying sub-fund fees, but excludes investment management, platform and transaction fees.

BlackRock

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BlackRock selects candidate ETFs by reference to:

- Low cost
- Targeted exposures (country, sectors and factors)
- High liquidity
- Operational efficiency
- Tight tracking error

iShares Core MSCI World ex Australia ESG ETF

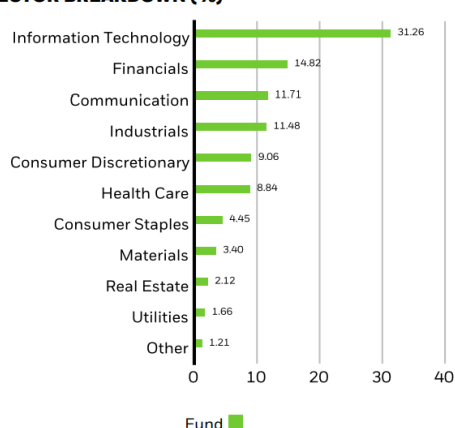
iShares
by BlackRock

Top 10 Holdings

NVIDIA CORP	11.13%
MICROSOFT CORP	6.61%
ALPHABET INC CLASS A	5.14%
ALPHABET INC CLASS C	4.26%
TESLA INC	2.48%
ELI LILLY	1.72%
ADVANCED MICRO DEVICES INC	1.32%
ASML HOLDING NV	1.28%
VISA INC CLASS A	1.28%
INTEL CORPORATION CORP	0.98%
Total of Portfolio	36.20%

Holdings are subject to change.

SECTOR BREAKDOWN (%)

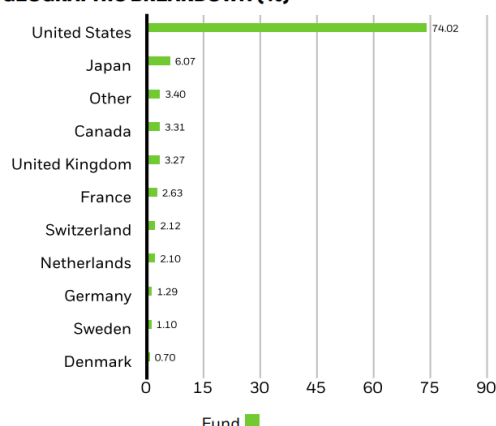


Allocation breakdowns shown above are subject to change. Total allocation percentages may not equal 100% due to rounding or omission of holdings of less than 1%. **Source:** BlackRock

TRADING INFORMATION

Exchange	ASX - All Markets
Ticker	IWLD
IRESS Ticker	IWLD.AXW
Bloomberg Ticker	IWLD
Listing Currency	AUD

GEOGRAPHIC BREAKDOWN (%)



Allocations are subject to change. Total allocation percentages may not equal 100% due to rounding or omission of holdings of less than 1%. **Source:** BlackRock

What's Important: Key observations:

- Despite the inclusion of other indices, equities indices always dominate holding 73% of total mortgage offset capital
- On a look-through basis 74% of equities exposure is to U.S. companies
- All candidate indices used in the reference asset portfolio are strongly correlated to the S&P Index

Appendix D: Long-Horizon Equity Hedging (SpiderRock)

This appendix sets out the continuous dynamic hedging program (Step 4 of the Payments & Risk Waterfall) designed and managed by SpiderRock, and the regime-aware framework used to size it.

Executive Summary

The EPM does not attempt to individually hedge all candidate indices used by BlackRock in the reference asset portfolios of each borrower's mortgage offset account.

Instead, it fully hedges only the S&P500 index because:

- The objective is to achieve equities-like returns with mortgage-like security
- Set-and-forget (until next 1, 3 or 5 year re-set) – hedging is not used as a strategic tool to play the markets and generate synthetic returns
- All candidate indices used by BlackRock are strongly correlated to the S&P index performance
- This correlation moves towards 1 (extremely high) in periods of economic downturn

Hedging is used as a risk management tool to:

- Protect offset capital
- Limit losses
- Further reduce reference asset portfolio volatility down to 10%

Key features of the S&P500:

- The S&P; 500 shows weak mean reversion over 30 years; long-horizon outcomes are dominated by drift uncertainty and regime shifts.
- Baseline probability of breaching an 80% real-wealth floor $\approx$ 16%.
- Left-tail risk is driven by persistent stagnation, not single crisis events.
- A combined static-replication sleeve + rules-based overlay reduces breach probability to 8–10%.
- Total expected hedge cost: 40–60 bps per year.
- Long-horizon equity risk is structural, not cyclical.

Why Index Mean Reversion Over 30 Years Is Weak

Individual stocks mean-revert; index mean-reversion is weak — this is the core modelling insight. Index composition changes (winners scale, losers exit), sector weights shift (tech → AI → next cycle), valuation anchors drift with real rates and margins, drift uncertainty compounds over decades, and structural breaks reset the "mean" repeatedly.

Modelling Framework — Regime-Switching Engine

The institutional standard for long-horizon equity modelling: GBM inside each regime, Markov switching across regimes, with volatility mean-reverting while returns do not. This produces realistic fat-tailed distributions. Drift & volatility bands (annualised):

Regime	Drift (μ)	Volatility (σ)
A: High-Growth / Low-Vol	6.5–8.0%	10–14%
B: Mid-Cycle / Normal	4.5–6.0%	14–18%
C: Stagnation / High-Vol	1.5–3.5%	18–25%
D: Crisis	-20% to -35% shock	35–55%

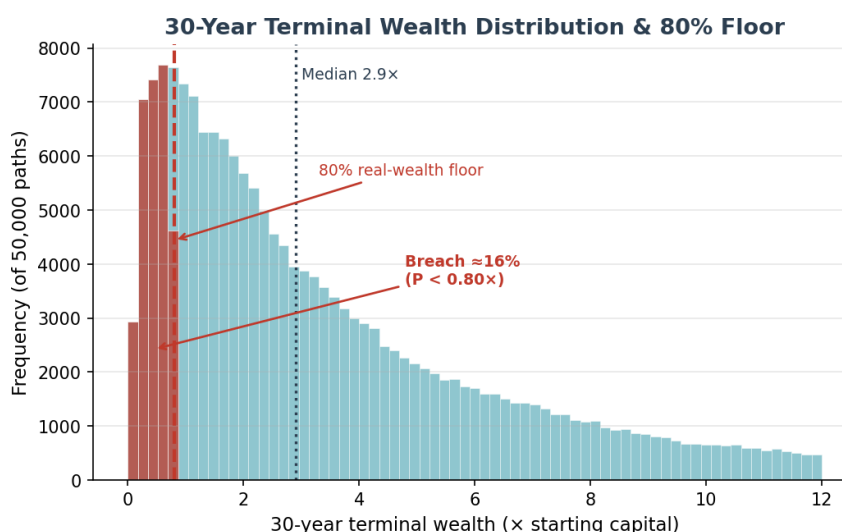
Regimes have 5–10 year half-lives. Stagnation (C) is the most persistent "bad" state; Crisis (D) is rare but sticky; High-Growth (A) is real but not dominant. Persistence is the key driver of long-run left-tail outcomes.

30-Year Terminal Wealth Distribution & Shortfall Risk

Percentile	Terminal wealth ($\times$)
5th	0.35 $\times$
25th	1.30 $\times$
Median	2.9 $\times$

Percentile	Terminal wealth ($\times$)
75th	6.27 $\times$
95th	16.7 $\times$

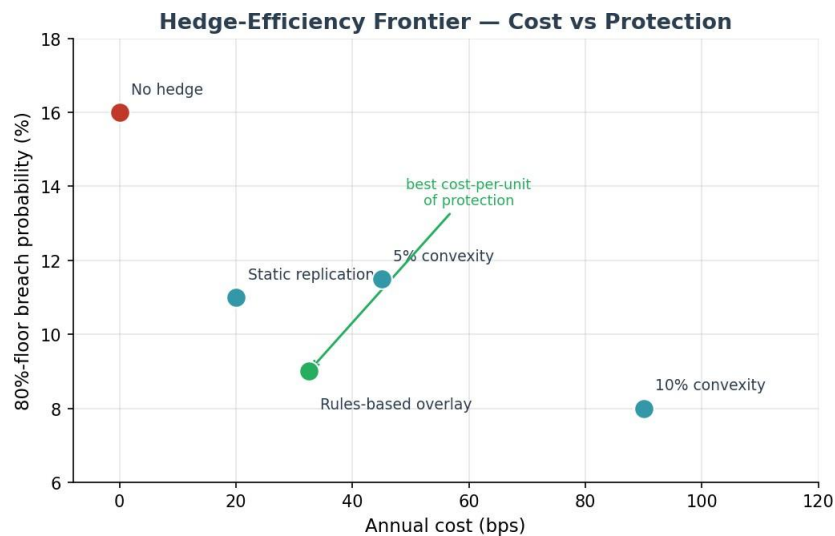
The distribution is wide because drift uncertainty dominates over 30 years. Shortfall risk against an 80% floor: $P(\text{terminal wealth} < 0.80) \approx 16\%$; $P(<1.0) \approx 19\%$; $P(<0.75) \approx 14\%$. The left tail is driven by persistent Regime C, not single-year crises — so crisis hedges alone are insufficient, which is what justifies the hedge program.



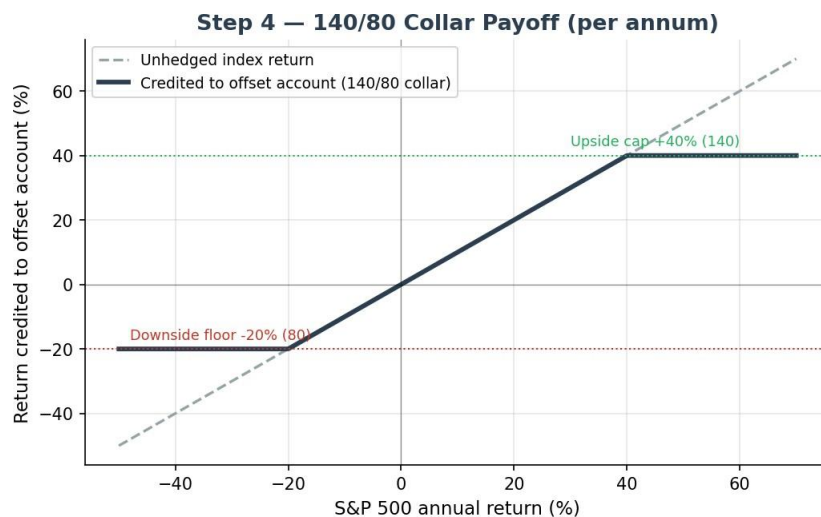
Hedge-Efficiency Comparison

Hedge structure	Annual cost	Breach probability	Reduction vs baseline
No hedge	0 bps	16%	—
5% convexity	35–55 bps	11–12%	4–5%
10% convexity	70–110 bps	7–9%	7–9%
Rules-based overlay	25–40 bps	8–10%	6–8%
Static replication	15–25 bps	10–12%	4–6%

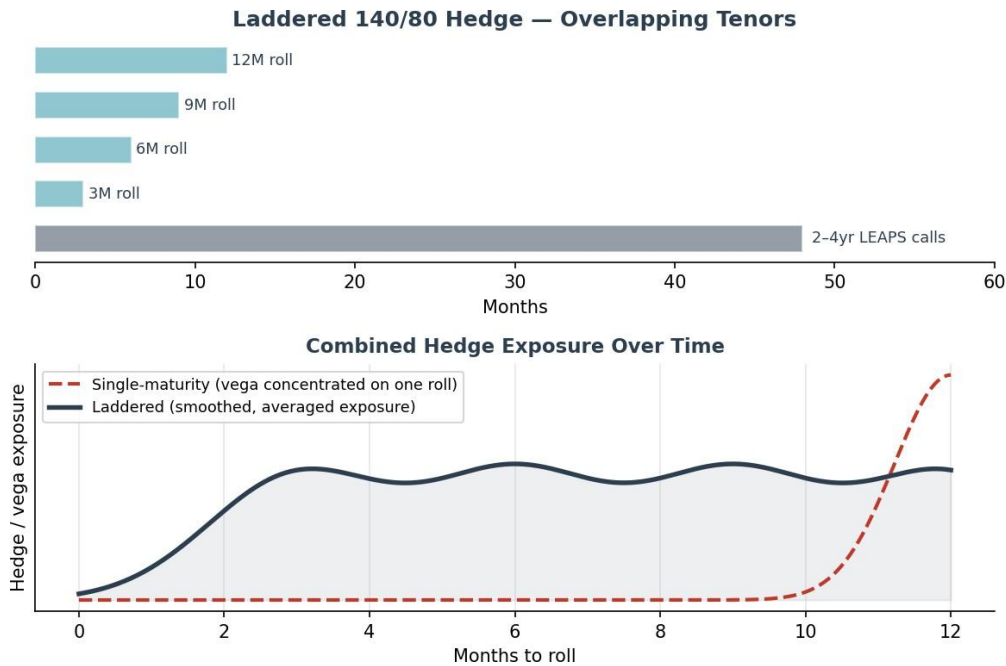
Rules-based overlays deliver the best cost-per-unit-of-protection.



How the Hedge Is Constructed — Laddered 140/80



The program uses a **laddered 140/80 collar** rather than a single-maturity structure. The payoff is identical (upside capped at 140%, downside floored at 80%, as shown above), but instead of all options expiring on one date, the hedge is built as four overlapping mini-hedges at 3M / 6M / 9M / 12M expiries (like roof tiles), with long-dated 2–4yr LEAPS calls across staggered tenors. A single-maturity hedge concentrates all vega, skew, gamma and execution risk on one roll date; the ladder spreads it.



- **Smooths vega** — not hostage to volatility on one day.
- **Reduces execution cost** — SpiderRock works smaller clips across expiries.
- **Reduces skew risk** — steep skew on one date affects only part of the hedge.
- **Reduces gamma cliffs** — no single expiry where the hedge "falls off".
- **Creates a regulator-friendly, systematic roll** — predictable, rules-based, low-variance.

Rules-Based Overlay & Static Replication Sleeve

Rules-based overlay. Activation needs 2 of 3 triggers: volatility ($VIX > 22$ or realised vol $> 18\%$), macro ($PMI < 50$ or credit spreads widen > 40 bps), trend (S&P; below 200-day MA or breadth $< 40\%$). Implementation: 5–10% notional OTM puts, 3–6 month tenor, 10–15% OTM strikes, rolled quarterly. Deactivation when $VIX < 18$, $PMI > 50$, trend recovers. Objective and regulator-friendly.

Static replication sleeve. 2–3% notional deep-OTM put spreads, 20–30% OTM strikes, 1-year tenor laddered, cost 15–25 bps. Always-on convexity that complements the overlay — the baseline protection layer.

***Combined outcome:** baseline breach probability 16% → with static sleeve 10–12% → with rules-based overlay 8–10%; total cost 40–60 bps. High efficiency, low drag, scalable.*

Implementation Pathway

- Phase 1: Deploy static replication sleeve.
- Phase 2: Add rules-based overlay.
- Phase 3: Quarterly regime review.
- Phase 4: Annual recalibration of drift/volatility bands.
- Phase 5: Integrate into product disclosure + risk governance.

***Conclusion:** A cost-efficient, regime-aware dynamic continuous hedge program reduces long-horizon shortfall risk by half, without compromising long-run equity participation.*